The supplied third Janko geometries without a distinguished action

A finite application of the accepted incidence certificates

Let $G = J_3$, of order $N = 50\,232\,960$, and let Γ_i, $1 \le i \le 21$, be the literal rank-three coset geometries in the acquired archive. Indices refer to its final assignment, which exchanges construction entries 8 and 9. The accepted eligibility certificates prove that all twenty-one are firm, residually connected, flag-transitive, residually weakly primitive, locally doubly transitive and satisfy IP_2. An isomorphism below may permute the types and need not preserve the specified image of G.

Proposition 1. *The supplied geometries have nineteen isomorphism classes. The only identifications are* $\Gamma_{13} \cong \Gamma_{14}$ *and* $\Gamma_{16} \cong \Gamma_{17}$.

Proof. Write $G_J = \bigcap_{j \in J} G_j$ for a base flag stabilizer, with $G_\varnothing = G$. Flag transitivity implies that the number of flags of type J is $N/|G_J|$: the orbit map sends a right coset $G_J g$ to the translate of the base flag, and equality of translates is precisely equality of these cosets. Therefore the multiset of all eight numbers $N/|G_J|$ is intrinsic to the incidence geometry. A type permutation merely permutes the eight subsets. Since N is the same for every input, the corresponding multiset of eight stabilizer orders is also an intrinsic separating invariant. This argument does not assume that an arbitrary isomorphism normalizes G.

The acquired intersection orders give sixteen distinct profiles. Their only nonsingleton fibres are

$$\{7, 8\}, \qquad \{9, 10, 11\}, \qquad \{13, 14\}, \qquad \{16, 17\}.$$

The saved whole-family automorphisms, conjugators and type permutations give actual incidence isomorphisms for the last two pairs: a map carrying the subgroup family to its target induces a well-defined bijection of cosets and preserves nonempty intersections in both directions.

The accepted common point-graph argument separates 7 from 8 and 9 from 11 even without a distinguished group action. It proves that any such geometry isomorphism must normalize the point action of G: the point graph has full automorphism group of order $2N$, whose derived subgroup is the distinguished G, and the other objects have distinct point shadows. The incompatible involution counts in the resulting parabolics then exclude both isomorphisms. We use that complete argument, not merely its invariant under group automorphisms.

Finally Γ_{10} satisfies IP, whereas Γ_9 and Γ_{11} fail WIP. Both properties are expressed solely through flags, incidences and shadows, so are preserved by arbitrary isomorphisms and by type permutations. The implication IP $\Rightarrow$ WIP separates 10 from both other members of its fibre. Thus the first repeated fibre splits into two classes and the second into three, while each of the last two is one class. The number is $16 + 1 + 2 = 19$. □

Source use and the scope of the count

The seventeen diagram labels of the acquired Janko supplement have twenty-one printed occurrences. Labels 3.7 and 3.8 each account for two distinct classes; labels 3.11 and 3.13 each account

for one class despite their two supplied occurrences. The other thirteen labels each account for one. In particular, counting the seventeen labels is not counting the nineteen supplied incidence classes.

L091 is D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics **33** (2005), 231–236. Its introduction expressly treats the Janko classification as previous work, citing reference [9]: D. Leemans, *The residually weakly primitive geometries of J_3*, Experiment. Math. **13** (2004), 429–433 (our historical identifier L097). Its Table 1 in Section 3 quotes comparison counts from those earlier classifications. The new HS classification and the two Hoffman–Singleton constructions are separate claims of L091, supported by their own accepted proofs.

The proposition proves the exact quotient of the supplied Janko roster and, unconditionally on any global census, the existence of at least nineteen eligible rank-three geometries. It does not enumerate all Janko subgroup families. To conclude that nineteen is the global rank-three total, one must retain the earlier classification's coverage premise, in the explicit form that every eligible geometry occurs in this roster. To interpret all nineteen as nontruncations, one must also retain its absence-of-higher-rank premise. These are borrowed classification inputs, not conclusions of the proposition; the old numerical value seventeen is not imported as a premise. In particular no claim of an independently reproved global Janko census is made.

The full text of L097 is unavailable in the fixed accessible catalogue, and L097 is outside the active 105 mathematical originals. The acquired thirteen-page author supplement supplies auxiliary data and does not change that status. L096, whose supplied group families and intersection certificates are reused here, is an active original with an accepted corrected principal recovery. Neither use awards a second original or recovery credit to L097. The Janko supplement's other diagram metrics, abstract subgroup names and full automorphism groups are not new assertions of L091 and need not be reclassified to audit its comparison table. Existing positive certificates for some of them remain available.

The read-only verifier acquires the accepted finite inputs, reconstructs every flag-count profile and this quotient, and records the historical source correspondence separately. It performs no new group search. The accepted GAP and Traces calculations retain their stated software premises. This packet makes no Lean, conjecture or compression claim.